

$K \xrightarrow{\text{key space}} \{0, \dots, 25\}; P_K[K=k] = 1/26 \text{ for each } k \in K$

Say we are given a following distribution over M :

$$P_M[M=a] = 0.7 \text{ and } P_M[M=z] = 0.3$$

What is the probability that the ciphertext is B?

There are only two ways to get the ciphertext B.

i) $M=a$ and $K=1$

ii) $M=z$ and $K=2$

By independence, we have

$$\begin{aligned} P_K[M=a \wedge K=1] &= P_K[M=a] \cdot P_K[K=1] \\ &= 0.7 \cdot (1/26) \end{aligned}$$

$$\begin{aligned} P_K[M=z \wedge K=2] &= P_K[M=z] \cdot P_K[K=2] \\ &= 0.3 \cdot (1/26) \end{aligned}$$

Using the Law of Total probability, we have

$$P[C=B] = 0.7/26 + 0.3/26 = 1/26$$

①

→ If multiple disjoint events can lead to the same outcome, we add the probabilities of each one.

What is the probability that the message 'a' was encrypted given that we observe ciphertext B.

Using Bayes Theorem, we have, $P[A|B] = \frac{P[B|A] \cdot P[A]}{P[B]}$

$$P_A[M=a | C=B] = \frac{P_A[C=B | M=a] \cdot P_A[M=a]}{P_A[C=B]} \rightarrow \text{from ①}$$

only possibility is $k=1$ $\swarrow$ $\frac{1/26 \cdot (0.7)}{1/26} \rightarrow$ Total probability of seeing ciphertext B

$$= 0.7$$

$\Rightarrow$ Even though the ciphertext was uniformly distributed, knowing that we observed B doesn't change our original belief about the message. Because the key is uniformly random.

$\therefore$ An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is PERFECTLY SECRET if for every probability distribution for $\mathcal{M}$, every $m \in \mathcal{M}$, and every ciphertext $c \in \mathcal{C}$ for which $P_A[C=c] > 0$:

$$P_A[M=m | C=c] = P_A[M=m] \quad (2.1)$$

Meaning: Seeing the ciphertext tells us nothing about the message.

In other words, perfectly secret means, Ciphertext is statistically independent of the message.

ANOTHER EXAMPLE:

Consider the shift cipher again with the following distribution over M .

$$P_M[M = \text{kim}] = 0.5 \quad P_M[M = \text{ann}] = 0.2 \quad P_M[M = \text{boo}] = 0.3$$

What is the probability that $C = \text{dgg}$?

The only way this given ciphertext can occur is if $M = \text{ann}$ and $K = 3$ and $M = \text{boo}$ and $K = 2$

$$\begin{aligned} \therefore P[C = \text{dgg}] &= 0.2 \cdot \left(\frac{1}{26}\right) + 0.3 \cdot \left(\frac{1}{26}\right) \\ &= \frac{0.5}{26} = \frac{1}{52} \end{aligned}$$

Now, what is the probability that ann was encrypted conditioned on observing the ciphertext dgg?

$$\begin{aligned} P[M = \text{ann} \mid C = \text{dgg}] &= \frac{P[C = \text{dgg} \mid M = \text{ann}] \cdot P[\text{ann}]}{P[\text{dgg}]} \\ &= \frac{\frac{1}{26} \cdot (0.2)}{\frac{1}{52}} = 0.4 \end{aligned}$$

This shows that the ciphertext is not perfectly secret.

Instead of reasoning about posterior probabilities $P_H[M=m | C=c]$, we use,

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A cipher is perfectly secure if and only if

$$P_H[Enc_k(m)=c] = P_H[Enc_k(m')=c]$$

$$\forall m, m' \in M, \forall c \in C$$

(2.2)

i.e., the probability of getting ciphertext c when encrypting m is the same as

(2.1) and (2.2) are actually equivalent.

PROOF:

$\Rightarrow$

Assume the scheme is perfectly secret. By definition, for every distribution on M , every $m \in M$ with

$P_H[M=m] > 0$ and every $c \in C$ with $P_H[C=c] > 0$, we have $P_H[M=m | C=c] = P_H[M=m]$ ----- (3)

By Bayes rule, $\forall m, c$,

$$P_H[M=m | C=c] \cdot P_H[C=c] = P_H[C=c | M=m] \cdot P_H[M=m] \text{ --- (4)}$$

Combine (3) and (4), we get,

$$(3) \Rightarrow P_M[M=m] \cdot P_C[C=c] = P_C[C=c|M=m] \cdot P_M[M=m]$$

Cancel $P_M[M=m] > 0$, we get,

$$P_C[C=c] = P_C[C=c|M=m] \text{ --- (5)}$$

Let $C = \text{Enc}_K(M)$ be the ciphertext. Then by conditioning,

$$P_C[C=c|M=m] = P_C[\text{Enc}_K(m)=c] \text{ --- (6)}$$

From (5) and (6), we have

$$P_C[\text{Enc}_K(m)=c] = P_C[C=c] = P_C[\text{Enc}_K(m')=c] \\ \forall m, m', c.$$

Thus (2.2) holds.

⇐

Assume $P_C[\text{Enc}_K(m)=c] = P_C[\text{Enc}_K(m')=c] \forall m, m', c$

To prove, $P_C[M=m|C=c] = P_C[M=m] \forall m, c$
with $P_C[C=c] > 0$.

Fix a message m with $P_M[M=m] > 0$ and a ciphertext c with $P_C[C=c] > 0$. Define $P_c := P_C[\text{Enc}_K(m)=c] \text{ --- (7)}$

By our assumption and since k is chosen independently of M , we have

$$P_H[C=c | M=m'] = P_H[\text{Enc}_k(m')=c] = P_c \quad \forall m' \in M$$

③

Using law of total probability, we have,

$$P_H[C=c] = \sum_{m'} P_H[C=c | M=m'] P_H[M=m']$$

$$= \sum_{m'} P_c \cdot P_H[M=m']$$

$$= P_c \cdot \sum_{m'} P_H[M=m'] = P_c \cdot 1$$

→ Sum of all the probabilities of mutually exclusive exhaustive outcomes for M .

$$= P_c$$

Using Bayes theorem,

$$P_H[M=m | C=c] = \frac{P_H[C=c | M=m] P_H[M=m]}{P_H[C=c]} = \frac{P_c \cdot P_H[M=m]}{P_c}$$

$$= P_H[M=m].$$

Thus observing $C=c$ does not change the probability of any m . This is exactly perfect secrecy.